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Highly Durable Bidirectional Rotary Triboelectric Nanogenerator with a Self-Lubricating Texture and Self-Adapting Contact Synergy for Wearable Applications

Xin Zhang, Mengjiao Liu, Zhiying Zhang, Hui Min, Ci Wang, Guangkai Hu, Tonghui Yang, Shimin Luo, Bin Yu, Tao Huang,* Meifang Zhu, and Hao Yu**

Sliding-freestanding triboelectric nanogenerators (SF-TENGs) are desirable for application in wearable power sources; however, improving their durability is the primary challenge. Meanwhile, few studies focus on enhancing the service life of tribo-materials, especially from an anti-friction perspective during dry operation. Herein, for the first time, a surface-textured film with self-lubricating property is introduced into the SF-TENG as a tribo-material, which is obtained by the self-assembly of hollow SiO_2 microspheres (HSMs) close to a polydimethylsiloxane (PDMS) surface under vacuum conditions. The PDMS/HSMs film with micro-bump topography simultaneously reduces the dynamic coefficient of friction from 1.403 to 0.195 and increases the electrical output of SF-TENG by an order of magnitude. Subsequently, a textured film and self-adapting contact synergized bidirectional rotary TENG (TAB-TENG) is developed, and the superiorities of the soft flat rotator with bidirectional reciprocating rotation are systematically investigated. The obtained TAB-TENG exhibits a remarkable output stability and an outstanding mechanical durability over 350 000 cycles. Furthermore, a smart foot system for stepping energy harvesting and wireless walking states monitoring is realized. This study proposes a pioneering strategy for extending the lifetime of SF-TENGs and advances it toward practical wearable applications.

1. Introduction

In the era of the Internet of Things (IoT) and big data, large numbers of smart portable electronics are widely distributed and required to be driven in real time.[1] Considering everyday bodily activities of human beings contain considerable available power, it is accessible to collect it in situ with wearable devices as a sustainable supplementary power source.[2] As a burgeoning technology invented in 2012 with the advantages of low cost, simple fabrication, and material selection diversity,[3] triboelectric nanogenerators (TENGs) aim to effectively harness manifold high-entropy energy from the environment,[4] as well as biomechanical energy from human motion.[5] So far, benefiting from the verified highest output energy density in all kinds of TENGs,[6] rotary-freestanding TENG (RF-TENG) derived from the sliding-freestanding TENG (SF-TENG) has been widely regarded as a prospective power source candidate in recent years.[7] However, severe issue of material abrasion caused by in-plane high-frequency sliding friction arises which will greatly hinder its further development.

For the purpose of improving the durability and long-term stability of the RF-TENGs, it can be fulfilled by simply reducing the contact time or area during the working process, including the strategies of noncontact rotating,[8] rolling electrification,[9] and automatic contact–noncontact state switching.[10] However, to ensure a high-level output performance, it is essential to maintain sufficient contact intimacy between the stator and the rotator. In this context, the fixing of elastic blades to the rotator is commonly employed to balance the high surface charge density and robustness with the aid of low-friction soft-to-rigid contact.[11] Afterward, Chen et al. systematically discussed the design optimization of soft-contact RF-TENGs for achieving a high-output performance, which can be regarded as a profound guidance.[12] Moreover, commercial animal fur can be used as a soft tribo-layer as well.[13] Although the present soft contact strategies can provide tight contact with low friction, tedious assembly procedure and requisite space occupation between the two disks will somehow impede the scalability and portability of RF-TENGs as wearable devices.

X. Zhang, M. Liu, G. Hu, T. Yang, S. Luo, B. Yu, T. Huang, M. Zhu, H. Yu
State Key Laboratory for Modification of Chemical Fibers and Polymer Materials
College of Materials Science and Engineering
Donghua University
Shanghai 201620, China
E-mail: yubin@dhu.edu.cn; ht@dhu.edu.cn; yh@dhu.edu.cn

Z. Zhang
School of Advanced Materials
Peking University Shenzhen Graduate School
Shenzhen 518055, China

H. Min
College of Information Science and Technology
Donghua University
Shanghai 201620, China

C. Wang
Shanghai Institute of Microsystem and Information Technology
Chinese Academy of Sciences
Shanghai 200050, China

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/smll.202300890

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Enhancing the service life of triboelectric pairs is an intrinsic way to overcome the limitation caused by material wear. As the mainstream approach, liquid lubricants (e.g., perfluoropolyether oil,[14] silicone oil,[15] squalane,[16] hexadecane,[17] polyalphaolefin,[18] synthetic grease,[19] and water-based graphene oxide solution[20]) have been introduced into the sliding interface to simultaneously decrease the frictional resistance and suppress the electrostatic breakdown effect, which is beneficial to the anti-wear property and the electric output. Nevertheless, excessive addition and uneven spreading of the liquid lubricant can significantly weaken the contact electrification between the two tribo-layers. In addition, during continuous device operation, the volume loss of the liquid lubricant increases. Considering the above points, it is, therefore, preferable to focus on modification of the tribo-materials to realize efficient operation under dry friction conditions.

Surface texturing is an intriguing method for reducing the coefficient of friction (COF). This is achieved by constructing specific micro/nano patterns on the surface, and can lead to a self-lubricating effect without the requirement for lubricant addition.[21] However, to the best of the authors' knowledge, this anti-friction surface has not yet been applied to TENGs. Furthermore, owing to the diminished surface elastic deformation resulting from a textured surface, it is necessary to further explore a matched surface contact mode and transmission structure of the RF-TENG exhibiting a lower friction force and a higher output.

Thus, we herein report an ingenious design for the preparation of a surface-textured film, wherein hollow SiO_2 microspheres (HSMs) are self-assembled close to the surface of a polydimethylsiloxane (PDMS) film under vacuum conditions to produce a micro-bump surface topography. The as-prepared PDMS/HSMs film can present self-lubricating property under dry friction conditions, achieving an ultra-low static and dynamic COF of 0.209 and 0.195, respectively. Compared with the pristine SF-TENG, the electrical output performance of the PDMS/HSMs-based SF-TENG shows nearly an order of magnitude enhancement. Subsequently, a textured film and self-adapting contact synergized bidirectional rotary TENG (TAB-TENG) is fabricated for the efficient harvesting of biomechanical energy. More importantly, high contact intimacy and low rotary resistance can be provided as well. The durability experiment indicates that even after 350 000 cycles there is almost no attenuation of the output voltage and the surface texture can be fully preserved. Finally, an integrated smart foot system with the TAB-TENG embedded is developed, demonstrating a considerable prospect of the highly durable TAB-TENG in the wearable applications of IoT and big data's era.

2. Results and Discussion

2.1. Fabrication and Characterization of the Surface-Textured Self-Lubricating PDMS/HSMs Film

The preparation process employed to obtain the PDMS/HSMs solution is schematically presented in Figure S1 (Supporting Information). Following the preparation of this solution, a surface-textured PDMS/HSMs film was fabricated via blade coating, vacuum treatment, and a heat curing process, as outlined in **Figure 1**a. It should be noted that the vacuum treatment step was the key to construct the textured surface. More specifically, owing to the vacuum-induced air pressure difference between the inside and outside of the HSMs, these particles floated upwards toward the PDMS surface and self-assembled into a close-packed structure. In contrast, the PDMS/HSMs film cured under atmospheric pressure exhibited a less distinct surface texture (Figure S2a, Supporting Information). Figure 1b shows a field-emission scanning electron microscopy (FE-SEM) image of fractured HSMs, which verifies their hollow structure. The FE-SEM image with lower magnification and the diameter distribution histogram of HSMs are exhibited as well (Figure S2b,c, Supporting Information). Compared with the pristine flat PDMS film (Figure 1c), the surface of the PDMS/HSMs film is fully textured with micro bumps and very uniform in each position (Figure 1d-i; Figure S3, Supporting Information), wherein the HSMs are stacked beneath the thin PDMS layer (Figure 1d-ii). Figure 1e presents the three-dimensional (3D) surface morphology of the PDMS/HSMs film, and the surface height information along the red dashed lines of the *x*- and *y*-axes are shown in Figure 1f. It can be seen that the scale of the surface texture fluctuated in the vertical direction within 7 μm, while the average surface roughness and root-mean-square surface roughness were 1.581 and 2.014 μm, respectively.

Surprisingly, the as-prepared surface-textured PDMS/HSMs film was found to possess an ultra-low COF with a self-lubricating property, which originated from its reduced surface elastic deformation and contact area.[21] To verify the surface friction behaviors, a static sliding friction test was carried out using a slope and an iron block (Figure S4, Supporting Information). The static COF was calculated from the relationship $\mu s_1 = H/L$, where H is the height at one end of the slope and L is the length of the slope when the iron block starts to slide. The derivation details are provided in Note S1 (Supporting Information). For the pristine PDMS film, the interfacial adhesion was so strong that no sliding of the iron block occurred at any inclination angle, even when the film was inverted by 180°. In stark contrast, the self-lubricating PDMS/HSMs film exhibited an extraordinarily low μs_1 value of 0.236 (Table S1, Supporting Information).

Owing to the inability of the static sliding friction test to measure the COF of the pristine PDMS film, a dynamic sliding friction test platform was constructed (Figure S5, Supporting Information). According to the GB 10006–88 standard, the dynamic friction force variation between the two material pairs during the uniform sliding process is shown in Figure 1g and Figure S6 (Supporting Information), and three self-lubricating PDMS/HSMs film samples were tested overall. The COF values of the pristine PDMS film and the self-lubricating PDMS/HSMs film during this process are shown in Figure 1h. Each material pair (i.e., PDMS–Cu and PDMS/HSMs–Cu) was tested in triplicate. Before the iron block began to slide, the friction force gradually increased, reaching the maximum static friction force (F_{s2}) when sliding commenced. At this point, the static COF (μs_2) was obtained from the normal force on the iron block (F_{N2}) and from F_{s2} (Note S2, Supporting Information). The μs_2 values of the pristine PDMS film and the self-lubricating PDMS/HSMs film were determined to be 1.802 and 0.209, respectively, which differ by almost an order of magnitude (Table S2, Supporting Information). In addition, the μs_1 and μs_2 values of the self-lubricating PDMS/HSMs film were comparable, both exhibiting low values.

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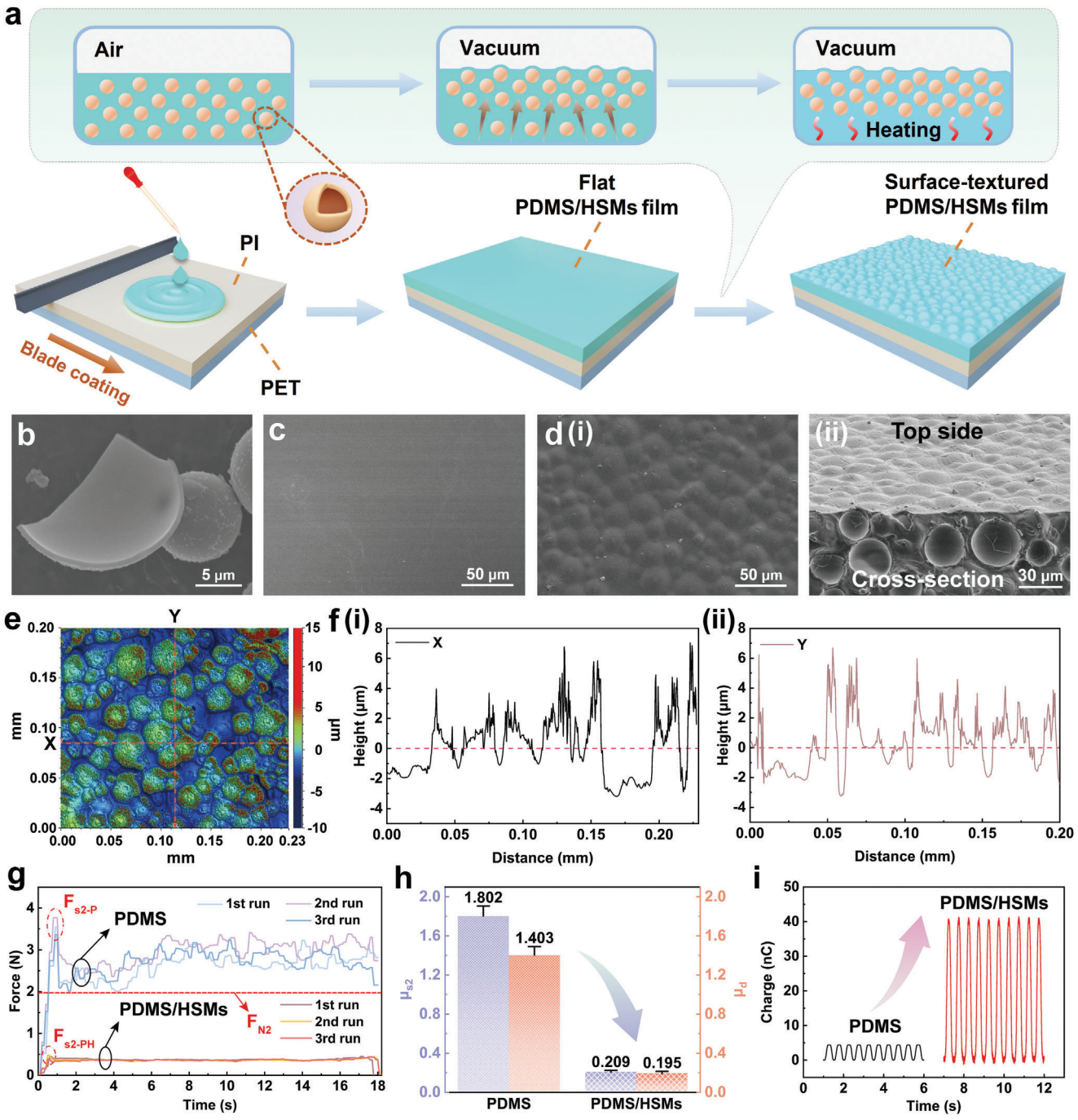


Figure 1. Preparation and characterization of the surface-textured self-lubricating PDMS/HSMs film. a) Schematic diagram of the fabrication process employed to obtain the PDMS/HSMs film. FE-SEM images of b) the fractured and intact HSMs, and c) the pristine PDMS film. d) FE-SEM images of the i) surface and ii) tilted cross-section of the surface-textured PDMS/HSMs film. e) 3D morphology image, and f) surface height information along the red dashed lines of i) the x-axis, and ii) the y-axis for the surface-textured PDMS/HSMs film. g) Dynamic friction force variation between the two friction material pairs during the uniform sliding process. h) Static and dynamic COFs obtained from the dynamic sliding friction test. i) Transferred charges of the two SF-TENGs with the pristine PDMS film and surface-textured self-lubricating PDMS/HSMs film as the electronegative tribo-layer.

Subsequently, as the iron block slid at a uniform velocity, the dynamic friction force varied with the sliding time, and the dynamic COF (μ_d) was obtained from the average dynamic friction force (F_{d2}) and from F_{N2} (Note S2, Supporting Information). As a result, it was found that the μ_d values of the pristine PDMS film and the self-lubricating PDMS/HSMs film were 1.403 and 0.195, respectively, with the latter representing an 87% reduction compared to that of the pristine film (Table S3, Supporting Information). Besides, the μs_2 and μ_d values of the PDMS/HSMs film cured under atmospheric pressure are 0.754 and 0.503,

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respectively, which is still at a relatively high level (Figure S7, Supporting Information). These results, therefore, demonstrate that excellent anti-friction property and an extremely low COF can be obtained under completely dry conditions via vacuum-induced micro-bump surface texture; this strategy shows great potential for application in SF-TENGs.

Thus, SF-TENGs were fabricated using the pristine PDMS film and the self-lubricating PDMS/HSMs film as electronegative tribo-layers, and the resulting output performances were tested at a reciprocating sliding frequency of 2 Hz (Figure S8, Supporting Information). The transferred charge, short-circuit current, and output voltage of the PDMS/HSMs-based SF-TENG were determined to be 40.9 nC, 0.5 μA, and 48.8 V, respectively, which are almost 10-fold higher than that of the pristine PDMS-based SF-TENG (Figure 1i; Figure S9, Supporting Information). This dramatically increased output may be ascribed to an improved contact intimacy and an enhanced sliding stability of the PDMS/HSMs–Cu triboelectric pair, which leads to a significantly increased contact electrification effect. Indeed, during the entire sliding process, the PDMS/HSMs–Cu triboelectric pair maintained full contact owing to the low COF, whereas the PDMS–Cu triboelectric pair showed discontinuous and unstable contact owing to the larger frictional resistance. Furthermore, this superior output performance could also be attributed to interfacial polarization between the HSMs and PDMS molecular chains, in addition to the excellent charge trapping effect of the HSMs electret, which is conducive to boosting the dielectric constant and charge retention ability of the as-fabricated PDMS/HSMs film.[22] However, when the weight ratio of HSMs in PDMS went too high (PDMS:HSMs = 3:1), the output performance of corresponding SF-TENG plummeted to a very low level because of much weaker triboelectric effect (Figure S10, Supporting Information), which was caused by both the broken surface with protruded HSMs and its scattered debris and limited PDMS component (Figure S11, Supporting Information). As an auxiliary proof, the dielectric constant of pristine PDMS film and self-lubricating PDMS/HSMs films with various weight ratios exhibited a similar trend as observed in the above-mentioned output results (Figure S12, Supporting Information). And the optimal weight ratio of PDMS and HSMs should be 5:1. Therefore, our surface-textured self-lubricating PDMS/HSMs film exhibits a significantly higher output performance on the premise of a distinctly reduced frictional resistance.

2.2. Structure, Working Mechanism, and Electrical Output Optimization of the TAB-TENG

To extend the utilization range of the surface-textured self-lubricating PDMS/HSMs film and harvest biomechanical energy in practical scenarios, a TAB-TENG was fabricated, as outlined in **Figure 2**a. The TAB-TENG device contained a power transmission unit and a rotary TENG. In terms of the power transmission unit, the inner gear train not only converts linear motion into a sliding TENG-friendly rotation, but it also delivers additional energy from the same external stimuli by amplifying the angular velocity and number of rotations.[23] The gear train consists of seven meshing gears (G1–G7), the details of which are listed in Table S4 (Supporting Information). For the purpose of this study, the gear ratio (N) was set at 30; this is defined as the ratio of the angular velocity of the final driving gear (G7) to that of the first driven gear (G1).[24] On this basis, through combining motion capture for the handle and the resilience of the tension spring, the obtained rotary TENG can be operated in the form of a periodic bidirectional rotation and accepted greater amounts of energy from the reciprocating downward and ascending movements with a limited short displacement. As for the rotary TENG, it consists of an electrode stator and a soft flat rotator. The electrode stator possesses two groups of complementary radial-arrayed sectors with a uniform central angle of ≈20° for each unit, in addition to a diameter of 70 mm (Figure 2b; Figure S13, Supporting Information). In contrast to previous designs that employ a rigid framework, we fabricated an assembly-free soft flat rotator with nine triboelectric sectors (Figure 2c). This soft flat rotator features a self-adapting contact property, and its light weight and low degree of inertia allow it to precisely meet the criteria of bidirectional transmission and rotation (as discussed in detail later). As shown in Figure 2d, a polyethylene terephthalate (PET) layer was used to provide the appropriate mechanical support, while a polyimide (PI) layer was added as the interlayer for charge storage. An illustration of the configuration and energy transmission in the rotary TENG is provided in Figure S14 (Supporting Information).

Taking two repeating sector–electrode units as an illustration, the basic working principle of the TAB-TENG is schematically illustrated in Figure 2e. Accompanied by the relative rotational motion between the Cu electrode and the PDMS/HSMs–PI layer, the electricity generation process can be universally divided into four states. The Cu electrode that makes first contact with the PDMS/HSMs layer is defined as electrode-1, and the adjacent electrode on the right is defined as electrode-2. In the initial state, the PDMS/HSMs layer completely overlaps with electrode-1. Due to the presence of a triboelectric series,[25] equal amounts of positive and negative charges are generated by the triboelectrification on electrode-1 and on the PDMS/HSMs layer, respectively. Benefiting from the incorporation of the PI layer, a portion of the negative charges become further trapped in it, resulting in a greater degree of triboelectric charge accumulation on the PDMS/HSMs–PI layer (Figure 2e-i).[26] Subsequently, when the PDMS/HSMs–PI layer slides towards electrode-2, positive charges flow from electrode-1 to electrode-2, generating a current via the external load until reaching the electrical equilibrium (Figure 2e-ii). The PDMS/HSMs layer adopts full contact with electrode-2, and all positive charges are transferred to electrode-2 (Figure 2e-iii). As the rotation continues, another PDMS/HSMs layer comes into contact with electrode-1 and the positive charges on electrode-2 return to electrode-1, thereby producing a reverse load current on the external circuit (Figure 2e-iv). For clarity of presentation, the ideal potential distributions of the two sector–electrode units under the aforementioned four states were visualized through a finite element simulation using COMSOL software, as shown in Figure 2f.

Corona charging technology was then adopted to rapidly increase the surface charge density, as shown in Figure S15 (Supporting Information). In this circumstance, the applied high voltage generated negative ionized charges, which were uniformly injected into the PDMS/HSMs film on the rotator surface by means of the electrostatic field between the needle tip and the electrode substrate.[27] Thus, the influence of PI interlayer

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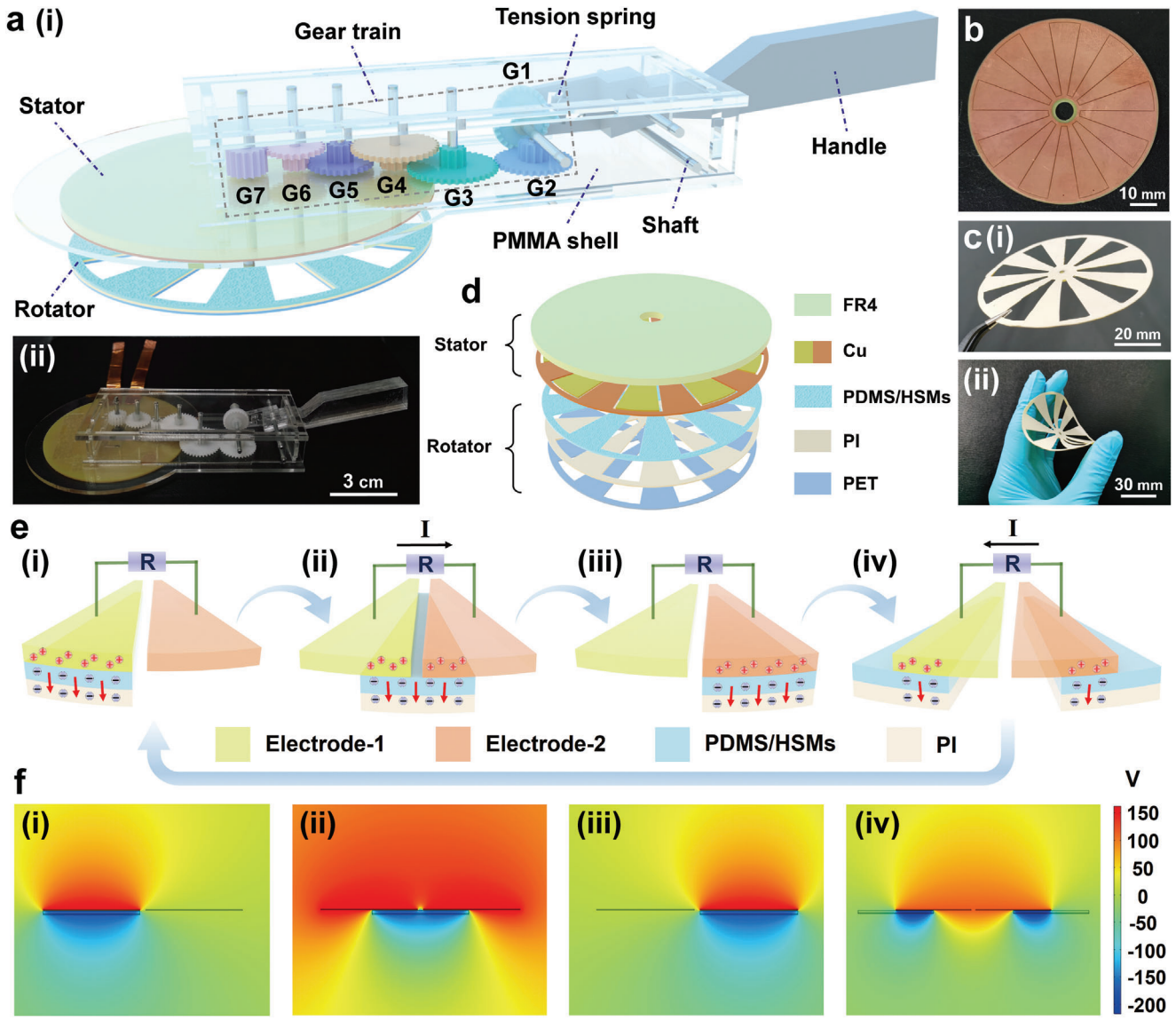


Figure 2. Architecture and working mechanism of the prepared TAB-TENG. a) i) Schematic diagram and ii) photograph of the TAB-TENG device. b) Photograph of the electrode stator. c) Photographs of the soft flat rotator exhibiting i) lightweight and ii) flexible properties. d) Schematic view of the structure and materials for the electrode stator and the soft flat rotator. e) Charge distribution and f) simulated potential distribution of i–iv) the four states existing during a single cycle of electricity generation at two sector–electrode units.

addition and corona charging treatment on the output performance of the prepared TAB-TENG was studied systematically, as shown in **Figure 3**a–c. More specifically, upon the incorporation of a PI interlayer, the maximum transferred charge, short-circuit current, and output voltage increased by 68, 65, and 65%, respectively. This was mainly attributed to the remarkable charge trap densities present in the deep trap states of the PI interlayer, which promote charge storage and enhance the electrical performance.[28] Meanwhile, PI interlayer can help retain surface charges and avoid dissipation for a long period of time (Figure S16, Supporting Information). In addition, compared to the nontreated TAB-TENG, corona charging resulted in the maximum transferred charge, short-circuit current, and output voltage increasing by 2.2, 2.1, and 2.4 times, respectively. In this case, not only a considerable surface charge can be produced during the corona charging process, but also the SiO_2 microspheres immobilized in the PDMS matrix and resulting interfaces are able to trap space charges.[29] A combination of these two strategies therefore resulted in the transferred charge density, short-circuit current density, and output voltage reaching maximum values of 11.4 μC m^{-2}, 5.1 mA m^{-2}, and 148 V, respectively, which are 3.5, 3.6, and 3.7 times higher than those of the initial TAB-TENG. It should also be noted here that all electrical output curves were obtained using a single pressing motion, which can be clearly divided into three stages (total duration ≈ 1.3 s), namely the pressing stage (I), the releasing stage (II), and the inertia

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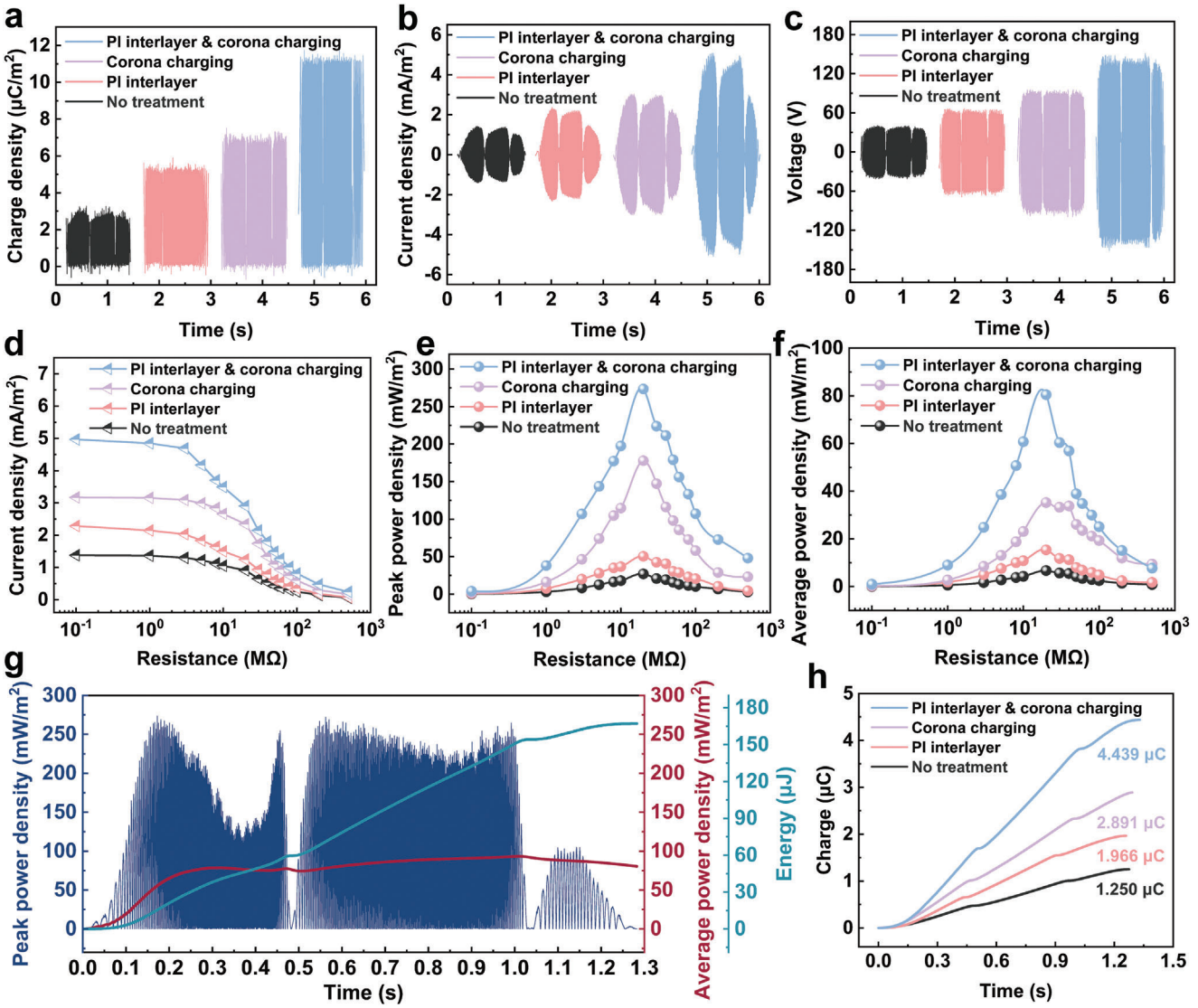


Figure 3. Influences of PI interlayer addition and corona charging treatment on the output performance of the prepared TAB-TENG. a) Transferred charges, b) short-circuit currents, and c) output voltages of the four TAB-TENGs during a single excitation. d) Peak currents, e) peak powers, and f) average powers of the four TAB-TENGs with respect to a load resistance of 0.1–500 MΩ. g) Real time variation curves of the peak power, the average power, and the energy generated from the optimized TAB-TENG at a load resistance of 20 MΩ in a single triggering process. h) Transferred charge quantities of the four TAB-TENGs as a function of time under a single external excitation.

driving stage (III) (Figure S17, Supporting Information). Unless otherwise specified, the reciprocating distance of the handle was 15 mm and the hand pressing force was 6 N.

The peak currents, peak powers, and average powers of the four TAB-TENGs are shown in Figure 3d–f, respectively, with respect to various load resistances. More specifically, for the optimal device, the peak current density (2.9 mA m^{-2}), maximum peak power density (274 mW m^{-2}), and maximum average power density (80 mW m^{-2}) were obtained at a load resistance of 20 MΩ. In addition, under a single external triggering process, the variation in peak power and average power of the optimized TAB-TENG over time can be observed in Figure 3g, along with the amount of generated electrical energy. As indicated, the average power density stabilized at ≈75 mW m^{-2} after an initial increase, while the electrical energy continuously increased to ≈0.17 mJ. Moreover, using the integral of the short-circuit current, the cumulative transferred charge quantity was derived as a function of time (Figure S18, Supporting Information), and the variation curves corresponding to the four TAB-TENGs are shown in Figure 3h. As indicated, the maximum value of 4.439 μC was obtained through a single hand pressing motion. All related equations are given in Note S3 (Supporting Information).

2.3. Superiorities of the TAB-TENG over the TRU-TENG

In the design of a RF-TENG with a power transmission unit, the unidirectional rotation mode is the mainstream strategy, and its

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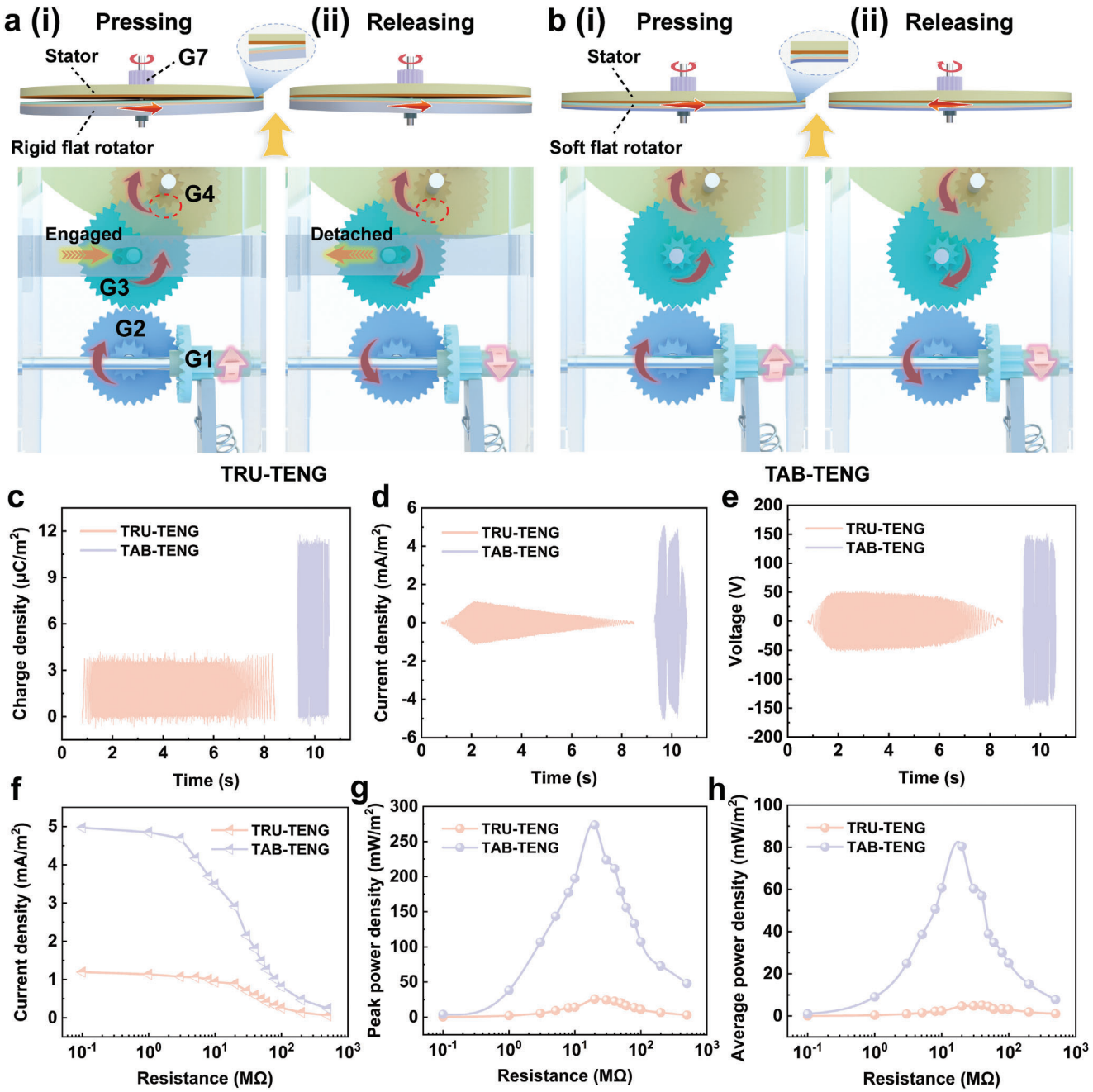


Figure 4. Comparison of the energy transmission, contact intimacy, and output performances of the TAB-TENG and TRU-TENG. Schematic diagram of the gear transmission mechanism and the contact intimacy between the stator and rotator of a) the TRU-TENG and b) the TAB-TENG in the i) pressing and ii) releasing stages. c) Transferred charges, d) short-circuit currents, and e) output voltages of the TRU-TENG and TAB-TENG in a single excitation. f) Peak currents, g) peak powers, and h) average powers of the TAB-TENG and TRU-TENG with respect to a load resistance of 0.1–500 MΩ.

matching with a rigid rotator is well accepted.[30] Surprisingly, we found that a bidirectional reciprocating rotation with a soft rotator can simplify the transmission unit while simultaneously reducing the device weight on the premise of retaining a high electrical output; this combination is preferred for biomechanical energy harvesting and wearable applications. Thus, a textured film and rigid contact combined with unidirectional rotary TENG (TRU-TENG) was fabricated for comparison with the TAB-TENG (Figure S19, Supporting Information). With the exception of the groove beam and the corresponding unfixed gear (G3), the overall structure of the TRU-TENG was identical to that of the TAB-TENG. It should be noted here that the soft flat rotator weighs <1 g, while the rigid flat rotator is 6.6 times heavier (Figure S20, Supporting Information). As shown in **Figure 4**a-i, when the handle is initially pressed, the energy is transmitted from G1 to G7, and G3 slides a short distance to the right to engage with G4.

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In the releasing stage, G3 slides a little to the left and detaches from G4, while the rigid flat rotator continues to rotate unidirectionally because of its own inertia; no energy transmission takes place at this point (Figure 4a-ii). Because of the imperfect contact originating from the poor deformation ability of the rigid flat rotator, any slight tilt or mismatch will dramatically increase the rotary resistance and the corresponding driving torque during the rotation process. In stark contrast, a lightweight soft flat rotator with negligible inertia (i.e., in the case of the TAB-TENG) can arbitrarily switch the rotational direction according to external excitation, which is preferable to match the bidirectional transmission mode in a noninterfering manner, thereby harnessing energy from both the pressing and releasing stages. More importantly, the soft flat rotator can further provide a self-adapting contact with the electrode stator, simultaneously reducing the rotary resistance and strengthening the contact intimacy (Figure 4b).

The differences in the energy transmission and contact intimacy between the TAB-TENG and TRU-TENG were clearly reflected in their resulting output performances. More specifically, when both TENGs contained a PI interlayer and were subjected to corona charging treatment, the maximum transferred charge density, short-circuit current density, and output voltage of the TRU-TENG was found to be 3.7 μC m^{-2}, 1.1 mA m^{-2}, and 50 V, respectively (Figure 4c–e), which are 68, 78, and 66% lower than the corresponding values for the TAB-TENG. These results indicate that a poor contact intimacy results in a small peak output. Similarly, when the peak current values under various load resistances were obtained, the maximum peak power density and average power density of the TRU-TENG were 25.6 and 4.7 mW m^{-2}, respectively, which are over an order of magnitude lower than those of the TAB-TENG (Figure 4f–h). Even though the total transferred charge quantity of the TRU-TENG is 4.449 μC under a single external excitation (Figure S21, Supporting Information), which is basically equal to that of the TAB-TENG (4.439 μC), the total transferred charge quantity of the TRU-TENG is mainly accumulated from a longer lasting time (i.e., 7–8 s) in one complete incentive process. Therefore, it will be more significantly affected when the TRU-TENG is applied in a more practical scenario (e.g., walking) that triggered by the continuous stimuli with shorter lasting time between each one, so it is necessary to further compare the continuous electrical outputs of the TAB-TENG and TRU-TENG using hand-driven frequencies of 0.5, 1.0, and 1.5 Hz. As shown in **Figure 5**a–c, the transferred charge density of the TRU-TENG was relatively consistent at these three frequencies, while the short-circuit current density and output voltage curves tended to become smoother as the frequency increased. In the case of the TAB-TENG, a time interval was observed between two independent electrical output curves at a frequency of 0.5 Hz. Upon increasing the frequency to 1.0 and 1.5 Hz, the TAB-TENG was able to continuously generate electricity, and the final inertia driving stage (i.e., Stage III) disappeared because the time interval between the two inputs was shorter than the normal duration of all three stages. In order to evaluate the electrical performances of TRU-TENG and TAB-TENG more comprehensively, their charging voltages at 10 μF capacitance with operating frequencies of 0.5, 1.0, and 1.5 Hz were determined, as shown in Figure 5d. It was found that the charging rate of the TAB-TENG at an operating frequency of 0.5 Hz was higher than that of the TRU-TENG at an operating frequency of 1.5 Hz. Moreover, this 10 μF capacitor could be rapidly charged to 9.8 V in 19 s by the TAB-TENG with an operating frequency of 1.5 Hz, which represents a charging rate of 5.16 μC s^{-1}, and demonstrates the outstanding superiority of the TAB-TENG over the TRU-TENG.

Due to the synergistic effect of the self-adapting property of the soft flat rotator and the self-lubricating property of the surface-textured PDMS/HSMs film, the TAB-TENG device exhibited an excellent durability. As shown in Figure 5e, almost no attenuation was observed in the output voltage under continuous measurement for 350 000 cycles, thereby implying a remarkable output stability. The mechanical durability of the surface-textured PDMS/HSMs film was also evaluated by comparing the degree of surface abrasion before and after 350 000 cycles. As shown in Figure 5f, no evident abrasion was present on the film surface after cycling, and the micro-bump topography was fully preserved. Consequently, the advantages of bidirectional energy harvesting, a tight contact intimacy, a higher electrical output, a lower rotary resistance, and a reduced degree of wear render TAB-TENG a superior candidate for biomechanical energy harvesting in practical scenarios.

2.4. An Integrated Smart Foot System for Stepping Energy Harvesting and Wireless Walking State Monitoring

Stepping is a common and indispensable motion for humans and can potentially provide a large amount of biomechanical energy. However, owing to its low frequency and limited displacement, this source of energy is rarely scavenged. At present, the majority of TENGs applied for foot applications are contact-separation TENGs that exhibit low, unstable, and intermittent electrical outputs. Despite the fact that RF-TENGs possess a high output energy density, their operating mode of rotation and rigorous demand of assembly constitute critical challenges in the context of wearable applications. Thus, as shown in **Figure 6**a, a TAB-TENG and book-shaped handle-driven TENG (BHD-TENG) were embedded in the insole of a homemade shoe to convert the reciprocating motion of the heel into electricity. The protruding part of the handle was set to accept the human stepping motion, and the employed TAB-TENG and BHD-TENG were both portable and detachable. With the aid of a power transmission unit and an assembly-free soft flat rotator, the TAB-TENG device was able to harness the stepping energy effectively, and importantly, it was easy to operate. As an auxiliary power supply, the BHD-TENG underwent a complete and continuous contact/separation motion with movement of the upper handle during the working cycle. A PI interlayer was also incorporated on both sides of the BHD-TENG to enhance the output performance, while polyamide-66 (PA-66) and nanoporous polytetrafluoroethylene (PTFE) were used as electropositive and electronegative tribo-materials, respectively (Figure S22, Supporting Information). The corresponding output performance is shown in Figure S23 (Supporting Information). Additionally, the TENG shoe was suitable for use (Figure 6b).

Triggered by the stepping motion, the embedded TAB-TENG was capable of illuminating 49 red light-emitting diodes (LEDs) connected in sequence with the shape of "DHU 70" (Figure 6c; Figure S24 and Movie S1, Supporting Information). The synchronous output voltage of the dual-channel TENG shoe is

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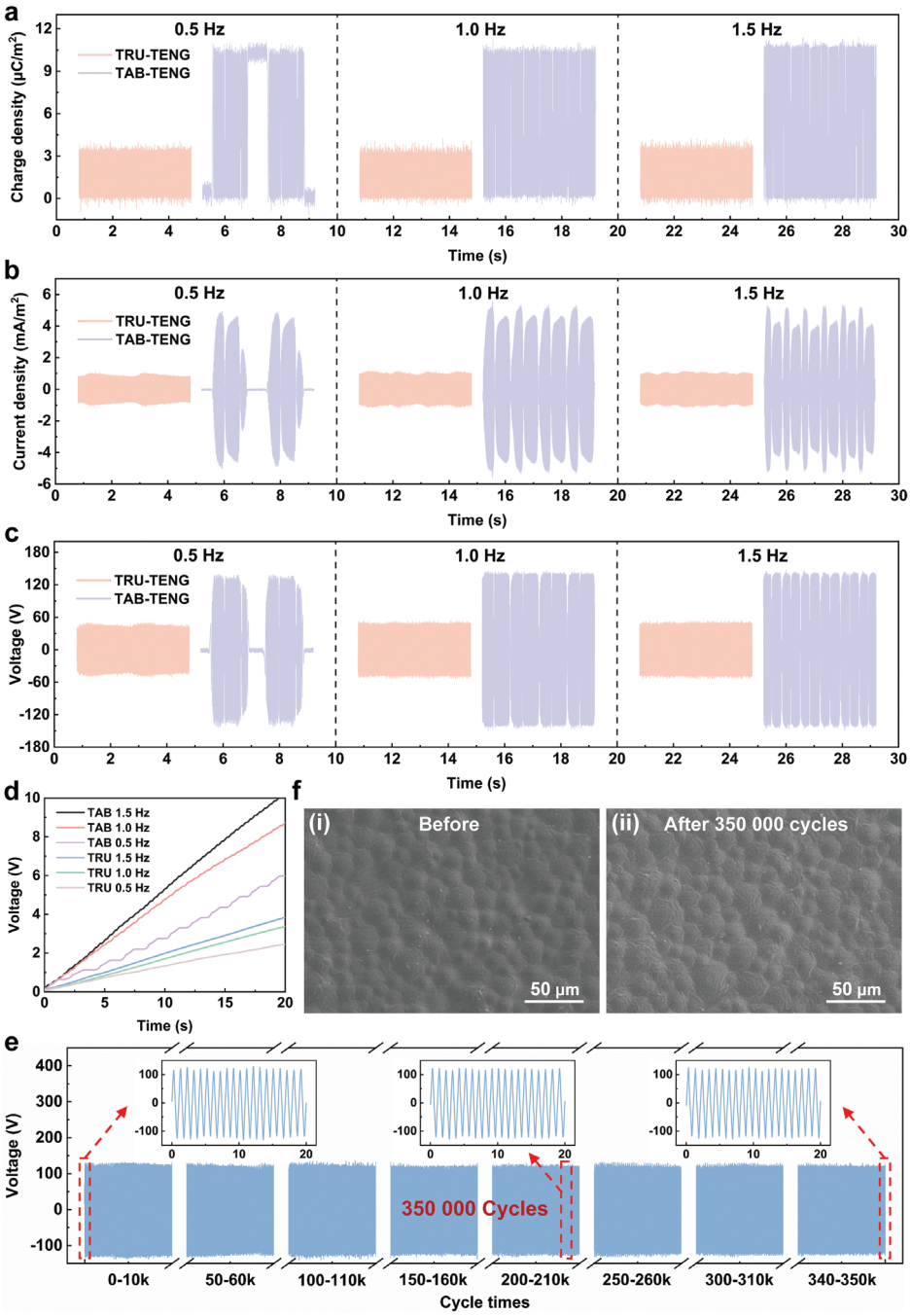


Figure 5. Comparison of the electrical outputs and charging performances of the TAB-TENG and TRU-TENG under different operating frequencies, and the superior durability of the TAB-TENG. a) Transferred charges, b) short-circuit currents, and c) output voltages of the TRU-TENG and TAB-TENG with operating frequencies of 0.5, 1.0, and 1.5 Hz for 4 s. d) Charging voltages of the TRU-TENG and TAB-TENG at 10 μF capacitance with operating frequencies of 0.5, 1.0, and 1.5 Hz. e) Stability of the output voltage under continuous measurement for 350 000 cycles. Inset: magnified views of the first few peaks, few peaks in the middle, and the last few peaks. f) Comparison of the FE-SEM images obtained for the surface-textured PDMS/HSMs film i) before and ii) after 350 000 cycles.

shown in Figure 6d, and the corresponding transferred charge and short-circuit current are presented in Figure S25 (Supporting Information). The charging performance of this device was then measured by connecting it to various capacitors via a full-wave bridge rectifier (Figure 6e), revealing a great feasibility for driving commercial electronics. As a result, an 8.5-inch color liquid crystal display (LCD) blackboard was successfully powered by the dual-channel TENG shoe containing a 47 μF capacitor, which could realize an overall handwriting and erasing function (Figure 6f-i,ii; Movie S2, Supporting Information). Furthermore, the charge–discharge curve recorded over the whole operating process is illustrated in Figure 6f-iii. Taking normal human

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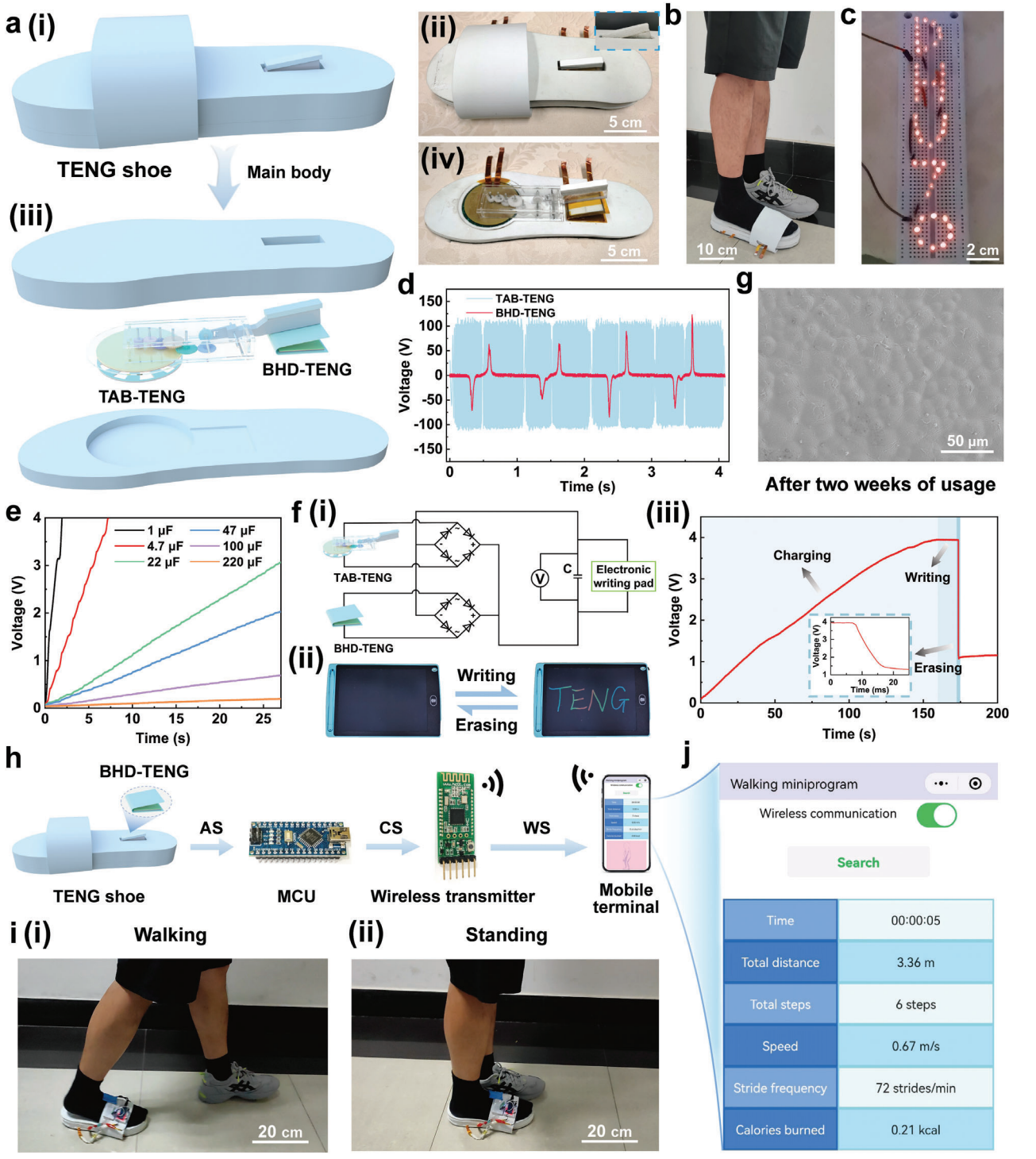


Figure 6. Demonstration of application of the integrated smart foot system for stepping energy harvesting and wireless walking state monitoring. a) i) Schematic diagram and ii) photograph of the TENG shoe. Inset: side view of the protruding part of the TAB-TENG handle. iii) Detailed schematic view and iv) photograph of the TENG shoe main body containing the embedded TAB-TENG and BHD-TENG. b) A user wearing the TENG shoe. c) Simultaneous illumination of 49 red LEDs connected in sequence with the shape of "DHU 70". d) Synchronous output voltages of the TAB-TENG and BHD-TENG triggered by the stepping motion. e) Charging voltages of the dual-channel TENG shoe at various capacitances under a 1 Hz stepping motion. f) i) Schematic circuit diagram, ii) before and after photographs of the screen, and iii) charge–discharge curve of a 47 μF commercial capacitor upon powering an 8.5-inch color LCD blackboard driven by the dual-channel TENG shoe at a stepping frequency of 1 Hz. Inset: enlarged view of the discharge curve when the handwriting on the LCD blackboard is erased. g) FE-SEM image of the surface-textured PDMS/HSMs film in embedded TAB-TENG after long-term practical applications. h) Signal flowchart of the wireless walking state monitoring system. i) User motions during real-time walking and standing status while wearing the integrated smart TENG shoe. j) User interface of the mini program.

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walking speed (i.e., ≈5 km h^{-1}) as a reference, the stepping frequency in the above measurement was ≈1 Hz. More importantly, after two weeks of consecutive tests equivalent to ≈3 h per day of stepping and walking, the embedded TAB-TENG exhibited superior durability as well as fully intact micro-bump morphology on the surface of PDMS/HSMs tribo-layer, as depicted in Figure 6g.

Finally, A wireless walking state monitoring system could be further fulfilled by the introduction of a microcontroller unit (MCU) and a wireless transmitter to the TENG shoe; a schematic circuit diagram and a digital photograph are shown in Figure S26 (Supporting Information). Thus, when a user wearing the integrated smart TENG shoe begins to walk, the BHD-TENG functions as a self-powered sensor, generating continuous alternating pulse outputs that synchronize with the user's footsteps as an analog signal (AS), and then transmits it to the MCU. By means of an analog-to-digital conversion module and further processing in the MCU, a digital control signal (CS) is produced and fed to the wireless transmitter. Subsequently, a wireless signal (WS) is established in the 2.4 GHz band by a wireless transmitter, then a mobile terminal receives the data through specific protocols and displays it on the screen in real time. The corresponding signal flowchart is illustrated in Figure 6h. For this purpose, a smartphone was used as the mobile terminal, and a customized mini program with an intuitive user interface was developed to visualize the walking progress (Figure 6i,j). A total of six indicators (time, total distance, total steps, speed, stride frequency, and calories burned) exist on the user interface, and these indicators are updated instantly upon variation of the walking state. Upon ceasing the walking motion, these indicators stop counting immediately, and the cartoon image changes from dynamic to static (Movie S3, Supporting Information). These results, therefore, indicate that an integrated smart foot system was successfully constructed, and that this system exhibits considerable potential for use in self-powered intelligent wearable applications.

3. Conclusion

In summary, we successfully fabricated a surface-textured PDMS/HSMs film with favorable self-lubricating property under dry friction conditions, which can reduce the static and dynamic COF from 1.802 to 0.209 and from 1.403 to 0.195, respectively. In addition, the electrical output of the PDMS/HSMs-based SF-TENG was more than an order of magnitude higher than that of the pristine SF-TENG, thereby indicating that the surface-textured PDMS/HSMs film could significantly decrease the frictional resistance and dramatically enhance the surface charge density. The TAB-TENG was further designed, wherein the optimal device containing a PI interlayer and treated by corona charging was able to generate a maximum transferred charge density, short-circuit current density, and output voltage of 11.4 μC m^{-2}, 5.1 mA m^{-2}, and 148 V, respectively. Importantly, in a single excitation, a maximum cumulative transferred charge quantity of 4.439 μC was generated. Furthermore, under a 20 MΩ load resistance, a maximum peak power density of 274 mW m^{-2} and a maximum average power density of 80 mW m^{-2} were obtained. Owing to the combination of the soft rotator and bidirectional reciprocating rotation, the TAB-TENG exhibited a superior tight contact intimacy, a higher electrical output, and a lower rotary resistance than the TRU-TENG. It was found that the self-lubricating texture and self-adapting contact synergy endowed the TAB-TENG with a strong robustness and an excellent durability, as reflected by a high degree of output voltage retention and no evident abrasion on the micro-bump surface under continuous measurement for 350 000 cycles. Finally, a dual-channel smart TENG shoe was employed as a stepping energy harvester and a wireless walking state sensor. Overall, this study pioneered a novel and facile approach to improve the durability of SF-TENGs and further pave the way for their usage in smart wearable applications.

4. Experimental Section

Materials: PDMS (Sylgard 184) was provided by Dow Corning Co., Ltd. *n*-Hexane was obtained from Sinopharm Chemical Reagent Co., Ltd. The HSMs (diameter: 10–30 μm) were supplied by Fanrui Composite Material Co., Ltd. The nanoporous PTFE film was purchased from Ningbo Dentik Fluor-Material Co., Ltd., while the PET film was obtained from 3 M Co., Ltd. All other commercial materials, including the ethylene vinyl acetate copolymer (EVA) sponge, the polymethyl methacrylate (PMMA) substrate, the PA-66 film, the PI tape, the foam tape, the Cu tape, and the conductive fabric tape (SY-NF35D-C, PET/Cu + Ni), were purchased from a local vendor.

Preparation of the PDMS/HSMs Film and the Pristine PDMS Film: The HSMs (1.1 g) were dispersed in *n*-hexane (3.85 g), and subsequently the PDMS base elastomer and the curing agent (5.5 g, 10:1, w/w) were added and dispersed under ultrasonication for 30 min to obtain a homogenous PDMS/HSMs solution. The mixed solution was poured onto the substrate and the blade-coated using a coating machine (AFA-IV, Xiandaihuanjing, China), followed by transferring to a vacuum oven. The coated film was then subjected to a vacuum of 0.1 MPa at 25 °C for 12 h, after which time, the PDMS/HSMs liquid film was cured at 80 °C for 3 h to obtain the PDMS/HSMs film. Similarly, the PDMS base elastomer and curing agent (6.6 g, 10:1, w/w) were added to *n*-hexane (3.85 g), and the dilute PDMS solution was poured onto the substrate and blade-coated, followed by curing at 80 °C for 3 h in an oven to obtain the pristine PDMS film.

Measurement of the COFs: For the static sliding friction test, Cu tape was attached to the slope, and an iron block (200 g, 63 × 63 mm^2) with a film sample (63 × 63 mm^2) adhered to the lower surface was placed at one end of the slope. The surface of the slope was gradually tilted to increase the inclination angle (θ), and the critical value of θ was recorded when the iron block started to slide. For the dynamic sliding friction test, the film sample bearing the PET substrate was fixed on an optical platform, and an iron block (200 g, 63 × 63 mm^2) with Cu tape (63 × 63 mm^2) attached to its lower surface was placed on the film sample. A linear motor (R-LP3, China) was introduced to control the uniform sliding velocity of the iron block (100 ± 10 mm min^{-1}), and the sliding distance was set at 30 mm. A digital force gauge (SH-5, rated load 5 N, HANDPI, China) was fixed on the PMMA cantilever beam connected to the linear motor, and its hook head was attached to the iron block to record the variation in the friction force.

Corona Charging: A 3 × 3 needle array was connected to a −15 kV high-voltage source and placed on top of the rotator with a vertical distance of 20 mm to generate a uniform high electrical field. The corona charging time is 2 min and the rectangular electrode beneath the rotator was grounded. The relative humidity and temperature were maintained at ≈20% and 25 °C, respectively.

Fabrication of the SF-TENG: For the electrode plate, a PMMA sheet (70 × 40 × 2 mm^3) was shaped using a laser cutter (Venus V-12 II, SYNRAD, USA) to serve as a substrate, and two rectangular Cu tapes (29 × 50 × 0.05 mm^3) were adhered to the substrate to act as electrodes. The entire electrode plate was attached to a custom-made linear motor with an in-plane reciprocating motion. For the freestanding tribo-layer, the double-sided foam tape (2 mm thick) and the film sample (29 × 45 mm^2, 100 μm-thick PET substrate layer, 150 μm-thick PDMS or PDMS/HSMs tribo-layer) were adhered to a long rectangular steel plate. The steel plate was fixed on a lab jack with adjustable height control.

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Fabrication of the TAB-TENG: The prepared TAB-TENG was composed of three parts, namely a bidirectional power transmission unit, an electrode stator, and a soft flat rotator. For the bidirectional power transmission unit, the sizes of the PMMA handle and shell were designed rationally and cut by a laser cutter, and the gear train was set to have seven gears engaged. Some commercial components, including tension springs, steel shafts, and silicone shaft sleeves, were also used to perfect the structure. It is worth mentioning that the stiffness of the tension spring is sufficiently high to ensure the effective resilience motion of the handle. For the electrode stator, it was fixed under the shell of the power transmission unit. The inner and outer diameters of the epoxy glass cloth laminate sheet substrate (FR-4) (1.6 mm thick) were 7 and 70 mm, respectively. The Cu layers (35 µm thick) with two groups of complementary radial-arrayed sectors (a total of 18 sectors, 0.3 mm gap distance) with the same central angle were coated on the substrate by means of printed circuit board techniques. The inner and outer diameters of the Cu electrode were 11.5 and 69 mm, respectively. Vias were drilled at the edge of the Cu electrode to connect the wires from the other side. For the soft flat rotator, the PI layer (55 µm thick) was adhered to the PET substrate (100 µm thick) as the interlayer, and the PDMS/HSMs layer (100 µm thick) on the top served as the negative triboelectric material. A radial-arrayed structure with nine evenly arranged hollowed-out sectors (central angle ≈20°) with an outer diameter of 70 mm and an inner diameter of 2 mm was shaped using a laser cutter.

Fabrication of the BHD-TENG: Initially, two pieces of EVA sponge with equal dimensions ($40 \times 30 \times 1$ mm^3) were adhered to either side of a rectangular hinged PET substrate ($100 \times 40 \times 0.1$ mm^3) that was folded in half lengthwise to ensure full contact between the two surfaces. The double-sided conductive fabric tape ($45 \times 35 \times 0.1$ mm^3) was adhered to the EVA sponge to act as the electrode. Subsequently, PI tape ($45 \times 35 \times 0.055$ mm^3) was attached to the conductive fabric to serve as the interlayer. Finally, PA-66 ($45 \times 35 \times 0.025$ mm^3) and a nanoporous PTFE film ($45 \times 35 \times 0.03$ mm^3) were adhered to the PI tape, acting as the positive and negative triboelectric materials, respectively.

Fabrication of the Dual-Channel TENG Shoe: The footwear upper and insole components were prepared from EVA sponge cut by a laser cutter. To properly embed the TAB-TENG and BHD-TENG, the EVA insole was prepared based on a multilayer structure with various hollowed-out patterns. More specifically, the bottom layer of the insole was composed of rigid polyvinyl chloride (PVC), which was cut by a laser cutter, and was employed to provide adequate deformation resistance for space maintenance during rotator spinning. The TAB-TENG, BHD-TENG, insole, and footwear upper components were assembled in sequence to obtain the dual-channel TENG shoe.

Characterization: FE-SEM (Hitachi SU8010, Japan) was used to observe the morphologies and microstructures of the HSMs, pristine PDMS films, and PDMS/HSMs films. A three-dimensional optical profiler (Contour GT-K 3D, Bruker, Germany) was used to detect the surface roughness values and 3D morphologies of the film samples. The dielectric property was measured by a broadband dielectric/impedance spectrometer (Concept 40, Navocontrol Technologies, Germany). The transferred charges and output currents of all TENGs, the voltages of charging the various capacitors, and the charge–discharge curves of driving the electronic devices, were measured using a programmable electrometer (Keithley model 6514, USA). An oscilloscope (Wavesurfer 104MXs-B, LeCroy, USA) with a probe impedance of 100 MΩ was used to detect the output voltages of all TENGs. Surface potentials of the PDMS/HSMs and PDMS/HSMs–PI films were determined using an electrostatic voltmeter (Trek 541A). The output measurements of the SF-TENGs were carried out under the reciprocating sliding motion of the Cu electrodes produced by a custom-made linear motor. The output measurements of the BHD-TENG were conducted under periodic compressive stress produced by a linear motor with a force sensor (SBT951, rated load 50 N, SIMBATOUCH, China).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

highly durable, rotary triboelectric nanogenerators, self-adapting contact, self-lubricating textures, wearable smart foot systems

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